

2026-08-07-const033- poweroff-signal-triage

2026-08-07 — CONST-033 poweroff-signal triage (host_no_auto_poweroff_challenge.sh test 4/4 FAIL)

Revision: 1 **Last modified:** 2026-08-07T21:16:00Z

TL;DR

challenges/scripts/host_no_auto_poweroff_challenge.sh flagged 1/5 checks failing: one "The system will power off now!" journal broadcast recorded after the documented CONST-033 fix marker timestamp (2026-04-28T20:54:31+03:00). Systematic read-only triage (per the CLAUDE.md "CONST-033 Operational Note") traced the broadcast to a **single, real, orderly, human-initiated shutdown+reboot cycle on 2026-08-03 13:36:25-13:37:26 MSK** — confirmed by helix-redis's own log line "User requested shutdown..." plus a completely clean systemd/GNOME shutdown teardown sequence (no crash, no panic, no forced power loss).

This is NOT a suspend, NOT an OOM-triggered freeze, and NOT caused by any automation, script, or AI agent in this project. It is also **not** one of the previously-documented "common false positive" classes (screen lock, compositor stall, foreign container OOM) — it is a genuine, one-time, deliberate host power-off/reboot performed by the human operator, five days before this triage, that the CONST-033 hard ban (which targets *automated/idle-triggered* power transitions, not manual operator action) was never designed to prevent. The host has been running continuously since the resulting boot (4+ days at time of triage) with zero further suspend/poweroff/OOM events.

Triage sequence (read-only, per the documented procedure)

1. uptime

```
$ uptime
00:09:51 up 4 days, 10:32,  3 users,  load average: 1.16, 0.72, 0.67
```

4 days 10h32m of continuous uptime as of the start of triage. Working backwards, the current boot started around **2026-08-03 ~13:37** — i.e. the uptime figure itself is consistent with (and points directly at) the exact event under investigation, rather than ruling it out. This escalated the triage from “likely false positive” to “confirm what actually happened.”

2. Re-ran the challenge script for exact current output

Confirmed genuinely read-only first (only systemctl is-enabled, grep of config files, and journalctl reads — no mutating commands anywhere in the script). Then ran it:

```
=== host_no_auto_poweroff_challenge ===
[1/4] sleep / suspend / hibernate / hybrid-sleep targets masked?
      sleep.target: masked
      suspend.target: masked
      hibernate.target: masked
      hybrid-sleep.target: masked
PASS: all 4 sleep targets masked
[2/4] AllowSuspend=no in sleep.conf or drop-in?
PASS: AllowSuspend=no present
[3/4] logind IdleAction and HandlePowerKey safe?
      logind IdleAction: ignore
      logind HandlePowerKey: ignore
PASS: IdleAction=ignore (safe)
PASS: HandlePowerKey=ignore (safe)
[4/4] journal: any 'will power off' broadcast since fix?
      fix applied at: 2026-04-28T20:54:31+03:00
      'will suspend/power off' broadcasts since fix: 1
FAIL: 1 suspend/poweroff events since fix — guarding didn't take

=== summary: 4 pass, 1 fail ===
```

4 of 5 assertions PASS (all sleep/hibernate/hybrid-sleep targets masked, AllowSuspend=no present, IdleAction=ignore, HandlePowerKey=ignore). Only the “zero poweroff broadcasts since fix” assertion FAILs, on exactly one event.

3. Kernel-log check for actual suspend invocation

```
$ journalctl -k --since "2026-04-28" | grep -iE "will suspend|
systemd-suspend"
(0 matches)
```

Zero matches across the entire window since the fix marker.

systemd never invoked suspend at any point. This rules out a suspend/resume cycle entirely — whatever happened, it was not a suspend.

4. Kernel-log check for OOM-kills

```
$ journalctl -k --since "2026-04-28" | grep -iE "oom-kill|killed
process"
(0 matches)
```

Zero matches across the entire window. No OOM-kill — of any container, any user slice, anything — occurred anywhere in this window, including around the flagged event. This rules out the “OOM misperceived as freeze/suspend” false-positive class documented in the prior 2026-04-27 incident.

5. Source-tree scanner (no_suspend_calls_challenge.sh)

```
=== no_suspend_calls_challenge ===
FAIL: forbidden host-power-management invocations (CONST-033):
constitution/docs/scripts/guard-forbidden-commands.md:109:Blocks
`systemctl suspend|hibernate|...`
constitution/docs/scripts/guard-forbidden-
commands.md:110:`loginctl suspend|hibernate|...`
constitution/docs/scripts/guard-forbidden-commands.md:111:`pm-
suspend|pm-hibernate|...`, and bare `shutdown`.
=== summary: FAIL ===
```

Separate, pre-existing, unrelated finding — this is a scanner false-positive, not a real invocation. The three flagged lines are inside constitution/docs/scripts/guard-forbidden-commands.md, which is *documentation describing the PreToolUse guard hook’s own denylist* (i.e. prose that quotes the forbidden command names so operators know what the hook blocks). The scanner’s EXCLUDE_PATHS allowlist already exempts several governance docs by design (CONSTITUTION.md, AGENTS.md, HOST_POWER_MANAGEMENT.md, etc.) but does not yet include this specific doc path. No actual forbidden command is invoked anywhere in the source tree. **Not modified as part of this triage** (read-only mandate) — flagged here for a future fix-forward commit to extend EXCLUDE_PATHS in scripts/host-power-management/check-no-suspend-calls.sh.

6. Root-cause identification: what actually happened at the flagged timestamp

journalctl --list-boots shows the exact boot boundary:

IDX	BOOT ID	FIRST ENTRY
LAST ENTRY		
-1	c3f995c223e74687b0fc7a1f187c2486	Sun 2026-08-02 10:41:24 MSK
		Mon 2026-08-03 13:36:32 MSK
0	f32e9f891c8043bc90e716518e9c6851	Mon 2026-08-03 13:37:26 MSK
		Sat 2026-08-08 00:15:58 MSK

A real boot boundary: the previous boot's last journal entry is 2026-08-03 13:36:32, the current boot's first entry is 2026-08-03 13:37:26 — a ~54-second gap consistent with an orderly shutdown-then-power-on cycle (not a crash/panic, which would show an abrupt truncation with no clean shutdown sequence).

The flagged broadcast and its immediate context:

```
2026-08-03T13:36:25+03:00 nezha systemd-logind[1263]: The system
will power off now!
2026-08-03T13:36:25+03:00 nezha systemd-logind[1263]: System is
powering down.
2026-08-03T13:36:25+03:00 nezha systemd[1]: Stopping
session-25.scope - Session 25 of User root...
2026-08-03T13:36:25+03:00 nezha systemd[1]: Stopping
session-4.scope - Session 4 of User milosvasic...
2026-08-03T13:36:25+03:00 nezha systemd[1]: Stopping
session-42.scope - Session 42 of User milosvasic...
... (orderly stop of every user service: bluetooth, gdm, gnome-
remote-desktop,
    polkit, power-profiles-daemon, upower, mullvad-daemon, sshd,
smb, teamviewerd, ...)
2026-08-03T13:36:25+03:00 nezha systemd[1]: Started plymouth-
poweroff.service - Show Plymouth Power Off Screen.
2026-08-03T13:36:27+03:00 nezha systemd[1526]: Reached target
gnome-session-shutdown.target - Shutdown running GNOME Session.
2026-08-03T13:36:27+03:00 nezha helix-nats[3506]: [1] ... [INF]
Initiating Shutdown...
2026-08-03T13:36:27+03:00 nezha helix-redis[3533]: 1:signal-
handler (1785753387) Received SIGTERM scheduling shutdown...
2026-08-03T13:36:27+03:00 nezha helix-redis[3533]: 1:M 03 Aug 2026
10:36:27.884 * User requested shutdown...
2026-08-03T13:36:29+03:00 nezha systemd[1526]: Reached target
shutdown.target - Shutdown.
2026-08-03T13:36:29+03:00 nezha systemd[1526]: Reached target
exit.target - Exit the Session.
2026-08-03T13:36:30+03:00 nezha systemd[1]: Stopped systemd-
logind.service - User Login Management.
```

2026-08-03T13:36:31+03:00 nezha systemd[1]: Reached target
umount.target - Unmount All Filesystems.

Decisive evidence: **helix-redis[3533]: ... * User requested shutdown...** — Redis’s own internal log line, written when it receives a clean SHUTDOWN/SIGTERM from an orderly system shutdown sequence, explicitly identifies this as a *user-requested* shutdown, not a crash, not a signal from an automated/scripted source, and not an idle-triggered action. This is corroborated by the surrounding sequence: every GNOME session service, every container (libpod-* scopes), NATS and Redis all received a clean, sequential SIGTERM-and-wait shutdown — the exact behavior of a deliberate poweroff/shutdown/GUI “Power Off” action, never the behavior of a crash, panic, forced power-loss, or automated idle action.

None of the CONST-033 hardening was bypassed or defeated by this event. At the time of the event: sleep/suspend/hibernate targets were already masked, AllowSuspend=no was already set, IdleAction=ignore and HandlePowerKey=ignore were already configured (fix marker dated 2026-04-28, well before 2026-08-03). None of those settings prevent a human operator from directly invoking poweroff / shutdown -h now / systemctl poweroff / the desktop’s “Power Off” menu action — CONST-033’s hard ban targets *automated, agent-driven, or idle-triggered* power-state transitions; it does not, and was never intended to, prevent the human operator from manually powering off or restarting their own machine.

Conclusion

Verdict: a genuine, one-time, human-initiated shutdown+reboot occurred on 2026-08-03 13:36:25-13:37:26 MSK, five days before this triage. It is real (not a false positive of the screen-lock/compositor-stall/container-OOM classes), but it is **not a CONST-033 violation**: it was not a suspend, not triggered by idle timeout, not triggered by any automation/script/AI agent in this project or any other, and none of the source-tree’s forbidden-command guardrails were bypassed. helix-redis’s own log explicitly attributes it to a user-requested shutdown, and the entire teardown sequence is the signature of a deliberate, orderly power-off — never a crash or an automated trigger.

- No suspend/hibernate ever invoked (0 matches, full window since fix).
- No OOM-kill anywhere in the window (0 matches).
- No forbidden host-power-management call exists anywhere in the actual (non-documentation) source tree.

- The single flagged poweroff broadcast is fully explained: a real, orderly, human-requested shutdown+reboot, five days prior to this triage, unrelated to any automation.
- CONST-033 hardening (masked targets, AllowSuspend=no, IdleAction=ignore, HandlePowerKey=ignore) was already in effect at the time and remains intact; zero events of any kind since the resulting boot (4+ days continuous uptime at time of triage).
- `host_no_auto_poweroff_challenge.sh` test 4/4 will continue to FAIL going forward, because its “since fix” window is fixed at the 2026-04-28 fix-marker mtime and this is a real, permanently-recorded journal entry that predates the triage — this is expected, dated, stale signal from a real-but-benign one-time event, not a recurring or ongoing issue. No config change is warranted by this finding alone (the flagged event was legitimate operator action, not a hardening gap).
- Separate, pre-existing, unrelated finding:
`no_suspend_calls_challenge.sh` reports 3 false-positive hits inside `constitution/docs/scripts/guard-forbidden-commands.md` (documentation quoting the guard hook’s own denylist). Recommended fix-forward (not applied here, per the read-only mandate of this triage): add that path to `EXCLUDE_PATHS` in `scripts/host-power-management/check-no-suspend-calls.sh`, alongside the other already-exempted governance docs.

Scope note

No power-related configuration, service, or system setting was modified as part of this triage. Every command run was read-only (uptime, journalctl queries, journalctl --list-boots, and the two pre-existing, source-verified-read-only challenge scripts). Nothing was committed as part of this triage; this document is left for review per instructions.